

MOBAN Chapter 5 — Comprehensive Study Guide

Wireless LAN (IEEE 802.11 / Wi-Fi)

PART 1: THEORY SUMMARY

1. Design Goals & Characteristics

Design goals of WLANs: - Global operation (available worldwide) - License-free operation (ISM bands) - Robust transmission technology - Easy to use, spontaneous cooperation - Low power for battery operation - Interoperability with existing wired networks - Transparency for applications - Security, privacy and safety

Advantages: - Very flexible within radio coverage - Ad Hoc networks possible without prior planning - Invisible design - More robust against disasters - No additional cost when adding users - Long equipment lifetime

Disadvantages: - Lower bandwidth, lower QoS, higher delay/jitter vs. wired (shared medium) - Lower safety, security, privacy - Many proprietary solutions; standards take time - National restrictions on radio operation

2. IEEE 802.11 / Wi-Fi Introduction

Wi-Fi = registered trademark of the Wi-Fi Alliance; tagline "Wireless Fidelity" - Wi-Fi Certified™ = interoperability label among vendors - Based on IEEE 802.11 family of standards

IEEE 802.11: - Specifies **PHY** and **MAC** layers - Same interface to higher layers (LLC) — interoperable with 802.x family - First version: IEEE Std 802.11-1997; Latest: IEEE Std 802.11-2024

Protocol stack position:

Application → TCP → IP → LLC → [802.11 MAC | 802.11 PHY]

3. Architecture

3.1 Infrastructure vs. Ad Hoc

Feature	Infrastructure	Ad Hoc
Control	Central (AP)	Distributed
Client complexity	Simple	More complex
Hop count	Single-hop to AP	Multi-hop possible

3.2 Key Terms

Term	Definition
STA (Station)	Terminal with radio contact and medium access
AP (Access Point)	Station integrated into WLAN and the Distribution System
BSS (Basic Service Set)	AP + stations within same radio coverage
ESS (Extended Service Set)	Multiple BSSs connected via DS = one logical network
DS (Distribution System)	Wired interconnection network between APs

Term	Definition
Portal	Bridge to other (wired) networks
IBSS (Independent BSS)	Ad hoc group of STAs in same coverage, same frequency

3.3 Protocol Architecture Layers

PMD (Physical Medium Dependent)	– modulation, coding
PLCP (Physical Layer Convergence Protocol)	– CCA, SAP
MAC (Medium Access Control)	– access, fragmentation, encryption
LLC (Logical Link Control)	– interface to higher layers
MAC Management	– association, roaming, auth, power mgmt
PHY Management	– channel selection, PHY MIB
Station Management	– coordinates all management functions

4. Physical Layer (PHY)

4.1 PHY Summary Table

Standard	PHY Type	Frequency	Data Rates	Year
802.11-1997	FHSS	2.4 GHz	1, 2 Mbps	1997
802.11-1997	DSSS	2.4 GHz	1, 2 Mbps	1997
802.11b	HR-DSSS	2.4 GHz	1, 2, 5.5, 11 Mbps	1999
802.11a	OFDM	5 GHz	6– 54 Mbps	1999
802.11g	ERP (OFDM)	2.4 GHz	1– 54 Mbps	2003
802.11n	HT PHY (MIMO)	2.4 & 5 GHz	6.5– 600 Mbps	2009

4.2 IEEE 802.11-1997: Basic Standard

Three original PHYs: - FHSS (Frequency Hopping Spread Spectrum) — 2.4 GHz - DSSS (Direct Sequence Spread Spectrum) — 2.4 GHz - Infrared (IR)

CCA (Clear Channel Assessment): indicates if the medium is idle (carrier sense)

Power limits: - USA: 1 W max transmit power - EU: 100 mW EIRP, min 1 mW - Japan: 10 mW/MHz

FHSS PHY: - 79 channels of 1 MHz each (US/Europe) - Minimum 2.5 hops/s - Modulation: 2-level GFSK (1 Mbps) or 4-level GFSK (2 Mbps)

DSSS PHY: - Barker code chipping sequence: +1, -1, +1, +1, -1, +1, +1, +1, -1, -1, -1 - 11 Mchips/s - Modulation: DBPSK (1 Mbps), DQPSK (2 Mbps)

4.3 CRC (Cyclic Redundancy Check)

Purpose: Detect transmission errors.

Concept: - Sender: multiply data D by 2^n , divide by generator polynomial, append remainder - Receiver: divide received frame by same polynomial → remainder = 0 means no error

Example (n=3, polynomial x^3+x+1 → pattern 1011):

Data D = 111001010 (9 bits)
Transmitted = 111001010 | 010 (frame = $D \cdot 2^3 + \text{remainder}$)
Receiver divides by 1011 → remainder = 0 → OK

Key formulas: - Frame size: $k + n$ bits (k = data, n = redundancy bits) - Polynomial of order $n \rightarrow$ pattern of $n+1$ bits

4.4 Scrambling

Purpose: DC blocking, spectral whitening, self-clocking.

Concept: - Sender: shift data back n bits, divide by polynomial $\rightarrow$ quotient = scrambled data - Receiver: multiply scrambled data by polynomial, shift forward n bits $\rightarrow$ original data

Note: Unlike CRC (where remainder is transmitted), with scrambling the **quotient** is transmitted — there is no remainder appended.

DSSS PHY scrambler polynomial: $s(z) = z^7 + z^4 + 1$

4.5 FHSS PHY Frame (PPDU)

```
| Sync (80b) | SFD (16b) | PLW (12b) | PSF (4b) | HEC (16b) | PSDU (var) |  
|-- PLCP Preamble (1Mbps) ---|--- PLCP Header (1Mbps) ---|-- payload --|
```

- **Sync:** 01010101... for bit sync and CCA
- **SFD:** frame sync pattern 0000110010111101
- **PLW:** payload length in bytes (<4096); includes 32-bit CRC of payload
- **PSF:** data rate (0000 = 1 Mbps, 0010 = 2 Mbps)
- **HEC:** 16-bit CRC of PLCP header using polynomial $x^{16}+x^{12}+x^5+1$

4.6 DSSS PHY Frame (PPDU)

```
| Sync (128b) | SFD (16b) | Signal (8b) | Service (8b) | Length (16b) | HEC (16b) | PSDU (var) |  
|--- PLCP Preamble (1Mbps) ---|----- PLCP Header (1Mbps) -----|-- payload --|
```

- **Sync:** scrambled 1-bits for sync, energy detection, frequency offset compensation
- **SFD:** 1111001110100000
- **Signal:** data rate (0x0A = 1 Mbps DBPSK; 0x14 = 2 Mbps DQPSK)
- **Length:** payload length in **microseconds** (<65536 μ s)
- All PHY bits are scrambled with $s(z) = z^7+z^4+1$

4.7 IEEE 802.11b: HR-DSSS (High Rate DSSS)

Key change: CCK (Complementary Code Keying) for 5.5 and 11 Mbps

Rates: 1 Mbps (DBPSK), 2 Mbps (DQPSK), 5.5 Mbps (CCK), 11 Mbps (CCK)

Channel plan (2.4 GHz ISM band): - 11–14 channels total (geography-dependent) - Channel spacing: 5 MHz - Channel bandwidth: 22 MHz - **3 non-overlapping channels:** - Europe: CH 1, 7, 13 - US/Canada: CH 1, 6, 11

Range: 300 m outdoor, 30 m indoor (max rate ~10 m indoor)

HR-DSSS frame: Same structure as DSSS but with long preamble of 192 μ s at 1 Mbps DBPSK; SFD is mirrored.

4.8 IEEE 802.11a: OFDM at 5 GHz

OFDM parameters: - 64 subcarriers total; 52 used (48 data + 4 pilot) - 12 virtual (guard) subcarriers - Subcarrier spacing: 312.5 kHz - Occupied bandwidth: 16.6 MHz (20 MHz channel) - Symbol duration: 4 μ s = 3.2 μ s data + 0.8 μ s guard interval - Symbol rate: 250,000 OFDM symbols/s

Data rates and modulation:

Rate	Modulation	Code Rate
6, 9 Mbps	BPSK	1/2, 3/4

Rate	Modulation	Code Rate
12, 18 Mbps	QPSK	1/2, 3/4
24, 36 Mbps	16-QAM	1/2, 3/4
48, 54 Mbps	64-QAM	2/3, 3/4

Mandatory: 6, 12, 24 Mbps

Frequency (Europe): 5.15–5.35 GHz and 5.725–5.825 GHz

Channel spacing: 20 MHz

Center frequency: $5000 + 5 \times \text{channel_number}$ [MHz]

802.11a PHY frame: - Preamble: 16 μ s (10 short + 2 long symbols) — for sync, channel estimation - Signal field: 1 OFDM symbol at 6 Mbps — carries rate, reserved, length (bytes), parity, tail - Data field: variable, at chosen rate

Range examples at 5 GHz: 54 Mbps $\leq$ 5 m, 36 Mbps $\leq$ 25 m, 18 Mbps $\leq$ 40 m, 12 Mbps $\leq$ 60 m

4.9 IEEE 802.11g: ERP (Extended Rate PHY) at 2.4 GHz

- Combines HR-DSSS (for 1, 2, 5.5, 11 Mbps) with OFDM (for 6–54 Mbps) in 2.4 GHz
- Same frame formats: Long/Short PPDU (from 802.11b) and ERP-OFDM PPDU (from 802.11a)
- Greater range than 802.11a due to lower frequency (300 m outdoor, 30 m indoor)
- Heavy interference on 2.4 GHz ISM band

4.10 IEEE 802.11n: HT PHY (High Throughput)

How 802.11n reaches 600 Mbps — step by step:

Enhancement	Change	Rate impact
More subcarriers	48 $\rightarrow$ 52 data subcarriers	54 $\rightarrow$ 58.5 Mbps
Better FEC	coding rate 3/4 $\rightarrow$ 5/6	58.5 $\rightarrow$ 65 Mbps
Short Guard Interval	800 ns $\rightarrow$ 400 ns (4 μ s $\rightarrow$ 3.6 μ s symbol)	65 $\rightarrow$ 72.2 Mbps
MIMO (4 $\times$ 4)	SISO $\rightarrow$ 4 spatial streams	72.2 $\rightarrow$ 288.9 Mbps
40 MHz channels	Bond two 20 MHz $\rightarrow$ 108 subcarriers	288.9 $\rightarrow$ 600 Mbps

Frequency: 2.4 GHz AND 5 GHz

Range: 70–200 m indoor, 200–800 m outdoor

PHY frame formats: - **Non-HT PPDU:** Same as 802.11a/g — backward compatible - **HT-Greenfield PPDU:** Optimized for 802.11n only; NOT backward compatible (causes collisions with 802.11a/g neighbors) - **HT-Mixed mode PPDU:** Combines non-HT preamble + new HT preamble; 802.11a/g clients can associate AND neighbors can decode preamble and back off — recommended for deployments with mixed clients

MCS (Modulation and Coding Scheme): single index encoding modulation + coding rate + stream count. Fast MCS feedback enables per-packet link adaptation.

5. MAC Layer

5.1 Traffic Services

Service	Type	How
Asynchronous Data	Mandatory	Best-effort, broadcast/multicast supported
Time-Bounded	Optional	PCF (Point Coordination Function) polling

5.2 Medium Access Methods

DFWMAC = Distributed Foundation Wireless MAC

Three access methods:

1. **DFWMAC-DCF CSMA/CA** (mandatory) — asynchronous, collision avoidance via random back-off + ACK
2. **DFWMAC-DCF with RTS/CTS** (optional) — solves hidden/exposed terminal; RTS reserves medium via NAV
3. **DFWMAC-PCF** (optional, infrastructure only) — AP polls STAs; supports time-bounded service

5.3 Inter-Frame Spaces (IFS) — Priority Mechanism

SIFS < PIFS < DIFS (shorter IFS = higher priority)

IFS	Priority	Usage	Formula
SIFS	Highest	ACK, CTS, polling response	PHY-specific
PIFS	Medium	PCF time-bounded service	SIFS + 1 slot
DIFS	Lowest	Asynchronous data (DCF)	SIFS + 2 slots

802.11a parameters: $T_{\text{SIFS}} = 16 \mu\text{s}$, $T_{\text{slot}} = 9 \mu\text{s}$ → $\text{DIFS} = 16 + 2 \times 9 = 34 \mu\text{s}$

5.4 DFWMAC-DCF CSMA/CA

Procedure: 1. Station senses medium (CCA) 2. If medium is **free for DIFS**: transmit immediately 3. If medium is **busy** (or after any successful transmission): wait DIFS + random back-off 4. Back-off timer **freezes** if another station occupies the medium (fairness) 5. Resume countdown when medium is free again; transmit when timer reaches 0

Contention Window (CW) — Exponential Back-off: - Low CW → many collisions; High CW → high delay - On collision: CW doubles - 802.11n example: $\text{CW}_{\text{min}} = 15$, $\text{CW}_{\text{max}} = 1023$ → CW sequence: 15, 31, 63, 127, 255, 511, 1023 - Back-off = random value in $[0, \text{CW}]$, unit = slot time

Unicast frames: - Sender waits DIFS (+ back-off if medium busy), then sends - Receiver replies after **SIFS** with ACK if CRC OK - No ACK → retransmit with new random back-off

Broadcast/multicast: No ACK

5.5 DFWMAC-DCF with RTS/CTS

Procedure: 1. Sender waits DIFS, sends **RTS** (with duration field = time needed for data + ACK + SIFSes) 2. Receiver replies after SIFS with **CTS** 3. Sender transmits data after SIFS 4. Receiver acknowledges with ACK after SIFS 5. All other stations hearing RTS or CTS set their **NAV** (Network Allocation Vector) → virtual carrier sense → defer access

Solves: - **Hidden terminal problem:** Station hidden from sender hears CTS → defers - **Exposed terminal problem:** Station exposed to sender hears RTS/CTS → can transmit to other destination

Duration field values: - In ACK: 0 (no more fragments) or remaining duration minus ACK time – SIFS - In RTS: time for CTS + data + ACK + $3 \times \text{SIFS}$ - In CTS: duration from RTS – time for CTS – SIFS

Fragmentation: - MAC-layer frame splitting to adapt to error rate (transparent to user) - Each fragment (except last) reserves medium for next fragment via duration field - Frame: RTS → CTS → frag1 → ACK1 → frag2 → ACK2 (with SIFS between each)

5.6 DFWMAC-PCF with Polling

Structure: Superframe = Contention-Free Period (CFP) + Contention Period (CP)

- AP = Point Coordinator; polls STAs in a list
- CFP starts with PIFS (higher priority than DIFS → AP wins contention)
- STAs set NAV to cover entire CFP

- If PCF only (no CP): equivalent to TDMA with TDD
- Extra overhead when nodes have nothing to send

Limits of PCF: No QoS guarantees, inflexible superframe length.

5.7 MAC Frame Format

| Frame Control (2B) | Duration/ID (2B) | Addr1 (6B) | Addr2 (6B) | Addr3 (6B) | Seq Control (2B) | Addr4 (0|6B)

Duration/ID: Medium occupation in μ s for NAV (<32768)

Sequence Control: Filters duplicate frames (due to lost ACKs)

CRC: 32-bit checksum (FCS)

Frame Control subfields:

Field	Bits	Meaning
Protocol version	2	0 = current protocol
Type	2	00=management, 01=control, 10=data, 11=reserved
Subtype	4	E.g., beacon (1000), RTS (1011), CTS (1100), data (0000)
To DS / From DS	1+1	Direction relative to Distribution System
More Frag	1	Another fragment follows
Retry	1	Retransmission frame
Power Mgmt	1	1 = power-save mode
More Data	1	AP has more data for sleeping STA
Protected Frame	1	Cryptographic encapsulation applied
Order	1	Process frames strictly in order

5.8 MAC Address Format

Four address fields: - **Address 1:** Physical receiver (filter here; ACK sent back to Address 2) - **Address 2:** Physical transmitter - **Address 3:** Logical receiver/sender/BSSID - **Address 4:** Logical sender (used only in DS-to-DS)

Scenario	To DS	From DS	Addr1	Addr2	Addr3	Addr4
Ad hoc	0	0	DA=RA	SA=TA	IBSSID	—
Infra: from AP	0	1	DA=RA	BSSID=TA	SA	—
Infra: to AP	1	0	BSSID=RA	SA=TA	DA	—
Infra: within DS	1	1	RA	TA	DA	SA

6. MAC Layer — IEEE 802.11e (QoS)

6.1 Why DCF/PCF is Not Enough

DCF limitations: No traffic differentiation, packet fairness (all stations same priority), best-effort only.

PCF limitations: Fixed superframe → inflexible, no mechanism to specify traffic requirements.

6.2 HCF: Hybrid Coordination Function (802.11e-2005)

Two new access methods: - **HCCA** (Hybrid Coordination Channel Access): PCF-like but more flexible CFP initiation - **EDCA** (Enhanced Distributed Channel Access): DCF-like but with priority differentiation

6.3 EDCA Access Categories

4 Access Categories (AC) mapped from 8 IEEE 802.1D User Priorities (UP):

AC	Traffic Type	Priority
AC_VO	Voice	Highest
AC_VI	Video	High
AC_BE	Best Effort	Low
AC_BK	Background	Lowest

6.4 EDCA Prioritization Mechanisms

1. AIFS (Arbitration Inter-Frame Space):

$$\text{AIFS[AC]} = \text{SIFS} + \text{AIFSN[AC]} \times T_{\text{slot}}$$

- Higher priority AC → smaller AIFSN[AC] → smaller AIFS → wins medium sooner
- Compare: DIFS = SIFS + 2 × T_slot (DCF)

2. Contention Window per AC: - CWmin[AC] and CWmax[AC] differ per AC - Higher priority → smaller CWmin, CWmax → smaller average backoff

3. TXOP (Transmission Opportunity): - Once a station wins the medium for an AC, it can send **multiple frames continuously** for a set duration - Only SIFS between successive frames within a TXOP

Intra-station contention: Each station runs 4 independent back-off engines (one per AC); internal winner competes on the wireless medium.

6.5 EDCA Limitations

- Still based on statistical fairness — no hard guarantees
- Streams of same priority still compete
- Performance varies with load and device implementation
- **Admission Control** needed for hard guarantees

6.6 Block Acknowledgment (Block ACK)

Introduced in 802.11e, critical for 802.11n:

- Sender sets ACK Policy = "Block ACK" in QoS Control field
- Multiple frames sent in a TXOP with only SIFS between them (no individual ACKs)
- Sender sends **BlockAckReq** after TXOP
- Receiver sends single **BlockAck** with bitmap indicating status of up to 64 MSDUs

Immediate vs. Delayed Block ACK: - Immediate: BlockAck sent after SIFS - Delayed: BlockAck sent after DIFS + back-off (for complex processing)

802.11e MAC frame addition: QoS Control field (2 bytes) appended when QoS subfield in Subtype = 1.

7. MAC Layer — IEEE 802.11n (Frame Aggregation)

Goal: Reduce per-frame overhead (preamble, DIFS, back-off, ACK) by sending more data per transmission cycle.

Two aggregation types:

	A-MSDU	A-MPDU
What is aggregated	Multiple MSDUs into one MPDU	Multiple MPDUs in one PPDU

	A-MSDU	A-MPDU
Max frame size	7935 bytes (vs. 2304 standard)	64535 bytes (vs. 4095)
ACK	Single ACK	Block ACK (no prior BlockAckReq needed)
Error handling	Lose all on error	Only retransmit failed MPDUs

8. MAC Management

8.1 Synchronization

Infrastructure: AP periodically broadcasts **beacon frames** containing: - Timestamp (real time!) — stations adjust local clock - Purpose: power management, PCF super-frame prediction, FHSS sequence sync

Ad hoc: Each node sends beacons; back-off applies to beacons; collisions possible → beacon lost.

Beacon interval: Typically 100 ms; delayed if medium busy (AP waits for idle medium).

8.2 Power Management

Goal: Switch off transceiver when not needed.

States: Sleep ↔ Awake

TSF (Timing Synchronization Function): Ensures all stations wake up at the same time.

Infrastructure mode (AP manages): - **TIM (Traffic Indication Map):** Beacon field listing STAs with buffered unicast frames - **DTIM (Delivery TIM):** Announces buffered broadcast/multicast frames - STA wakes at TIM interval, checks TIM, retrieves data if indicated, sleeps again - Trade-off: longer TIM interval → more power saving, but higher delay

Ad hoc mode: - **ATIM (Ad hoc TIM):** Stations announce buffered frames to receivers - More complex: no central AP, all stations buffer, collision of ATIMs possible

8.3 Roaming (Re-association)

Trigger: No or bad connection (RSSI below threshold)

Steps: 1. **Scanning:** Passive (listen for beacons) or Active (send Probe Request, receive Probe Response) 2. **Re-association Request:** STA sends to new AP 3. **Re-association Response:** AP accepts → STA joins 4. AP informs DS → DS updates location database → old AP releases resources

Note: No universal roaming solution across all vendors today.

9. Further Developments (802.11ax / Wi-Fi 6)

9.1 Paradigm Shift: Efficiency over Speed

Earlier 802.11 revisions focused on raw speed (higher PHY rate). **802.11ax** (Wi-Fi 6) targets **high efficiency** in dense deployments.

Four pillars of Wi-Fi 6: 1. UL/DL OFDMA 2. UL/DL MU-MIMO 3. BSS Coloring 4. Other enhancements (1024-QAM, TWT, etc.)

9.2 OFDMA (Orthogonal Frequency Division Multiple Access)

Concept: Partition the frequency-time space into Resource Units (RUs) and assign different RUs to different clients simultaneously.

802.11ac: One client uses the full channel per TXOP (contention per client).

802.11ax: AP coordinates multiple clients in one TXOP using sub-channel RUs.

RU sizes in 20 MHz: 26, 52, 106, 242 subcarriers

In 80 MHz: up to 996-subcarrier RU

Benefits: - Lower latency for short frames (don't wait for full channel) - Both uplink AND downlink - Single contention for multiple clients → reduces overhead

9.3 MU-MIMO (Multi-User MIMO)

Concept: AP uses spatial diversity to transmit to/from multiple clients simultaneously.

	802.11ac	802.11ax
Downlink MU-MIMO	Yes (up to 4 streams)	Yes (up to 8×8)
Uplink MU-MIMO	No	Yes
Max streams	4 clients	8 streams (Wave 2)

MU-MIMO + OFDMA: Combined in 802.11ax for RUs ≥ 106 subcarriers.

9.4 BSS Coloring

Problem: Overlapping BSSs (OBSS) → stations far from each other back off unnecessarily, wasting airtime.

Traditional CCA: If signal > SD threshold → backoff (regardless of which BSS it belongs to).

Solution — BSS Color: - 6-bit BSS identifier appended to PHY and MAC frames - Two SD thresholds: - **Same color:** Standard threshold — backoff (it's a neighbor in same BSS, collision possible) - **Different color:** Higher threshold — can transmit simultaneously (spatial reuse)

Result: Dense deployments with multiple APs on same channel can transmit simultaneously → less contention overhead.

9.5 Wi-Fi 6E

- Extends 802.11ax into the **6 GHz band** (5945–6425 MHz in Europe)
- Adds approximately 480 MHz of new spectrum
- Less congested than 2.4 GHz and 5 GHz

10. WLAN Throughput Calculation

10.1 Fundamental Formula

Throughput = Useful data bytes / Cycle duration

Useful data = IP payload (not LLC, MAC, or PHY headers)

10.2 Frame Encapsulation Stack (from exercise slides)

Application data
+ App header
+ TCP/UDP header → APP packet
+ IP header → TCP/UDP packet
+ LLC header (8B) → IP packet
+ MAC header + CRC → LLC packet (MSDU) → MPDU
+ PHY header + tail + padding → MAC Frame (PSDU) → PPDU

10.3 One Cycle: CSMA/CA (No RTS/CTS)

$$\text{Cycle} = \text{DIFS} + \text{back-off} + T_{\text{PHY_hdr}} + T_{\text{MAC_hdr}} + T_{\text{LLC_hdr}} + T_{\text{data}} + \text{SIFS} + T_{\text{PHY_hdr_ACK}} + T_{\text{ACK}}$$

Where: - $\text{DIFS} = \text{SIFS} + 2 \times T_{\text{slot}} - T_{\text{back-off}} = (\text{CWmin}/2) \times T_{\text{slot}}$ (average: $\text{CWmin} = 15 \rightarrow \text{avg back-off} = 7.5$ slots) - $T_{\text{xxx}} = (\text{frame size in bits}) / \text{PHY data rate} - \text{Preamble} + \text{signal field transmitted at 6 Mbps (base rate); payload at negotiated rate}$

10.4 One Cycle: CSMA/CA with RTS/CTS

$$\text{Cycle} = \text{DIFS} + \text{back-off} + T_{\text{RTS}} + \text{SIFS} + T_{\text{CTS}} + \text{SIFS} + T_{\text{data}} + \text{SIFS} + T_{\text{ACK}}$$

Effect of RTS/CTS: Adds two extra frames (RTS + CTS) per cycle $\rightarrow$ lower throughput for small packets but reduces collision probability in multi-station environments.

10.5 Fragmentation Impact

- Large payload fragmented into N fragments of size f_{max}
- Each fragment: RTS + CTS + data_fragment + ACK (+ SIFSes)
- Total cycle = $N \times \text{fragment_cycle}$
- More overhead per bit $\rightarrow$ lower throughput compared to no fragmentation

10.6 Key Numbers for 802.11a (from exercises)

Parameter	Value
T_{SIFS}	16 μs
T_{slot}	9 μs
T_{DIFS}	34 μs
CWmin	15
Max PHY rate	54 Mbps
RTS frame	20 bytes
CTS/ACK frame	14 bytes
MAC header	34 bytes
LLC header	8 bytes
PHY preamble+signal	16 μs + 4 μs = 20 μs

10.7 Exercise Solutions Summary

Exercise	Setup	Throughput
1	802.11a, CSMA/CA, 1500 B IP	30.8 Mbps
1	802.11a, CSMA/CA, 500 B IP	16.6 Mbps
2	802.11a, CSMA/CA + RTS/CTS, 1500 B	25.6 Mbps
3	802.11a, CSMA/CA + RTS/CTS + frag ≤ 500 B	16.7 Mbps
4	802.11a, CSMA/CA + RTS/CTS, 36 Mbps rate	20.4 Mbps
5	802.11n, CSMA/CA, 2 spatial streams, 1500 B	31.58 Mbps
6	802.11n, CSMA/CA, 2 SS, A-MPDU	64.29 Mbps
6	802.11n, CSMA/CA, 2 SS, A-MSDU	65.55 Mbps

Key observations: - RTS/CTS reduces throughput for a single sender (extra overhead), but becomes beneficial with hidden terminals - Fragmentation reduces throughput (more overhead per byte of useful data) - Frame aggregation (A-MPDU, A-MSDU) roughly doubles throughput by amortizing MAC overhead

PART 2: KEY FORMULAS REFERENCE

Formula	Expression
DIFS	$SIFS + 2 \times T_{slot}$
PIFS	$SIFS + 1 \times T_{slot}$
AIFS[AC]	$SIFS + AIFSN[AC] \times T_{slot}$
Average back-off	$(CW_{min} / 2) \times T_{slot}$
CW after k collisions	$\min(2^{k+1} - 1, CW_{max})$
Throughput (basic)	$IP_bytes / (DIFS + backoff + T_{data} + SIFS + T_{ACK})$
802.11a symbol duration	$4 \mu s (3.2 \mu s + 0.8 \mu s GI)$
802.11n short GI	$3.6 \mu s (3.2 \mu s + 0.4 \mu s GI)$
Throughput gain with aggregation N frames	$\approx N \times single_frame_gain / (overhead + N \times data)$

PART 3: CONCEPT COMPARISONS

DCF vs. PCF vs. EDCA

Feature	DCF	PCF	EDCA
Medium access	CSMA/CA	Polling by AP	Enhanced CSMA/CA
Priority	None	Fixed	Per traffic class
QoS guarantee	No	Soft	No (statistical)
Contention-free	No	Yes (CFP)	No
Ad hoc compatible	Yes	No	Yes

802.11a vs. 802.11b vs. 802.11g

	802.11b	802.11a	802.11g
Frequency	2.4 GHz	5 GHz	2.4 GHz
Max rate	11 Mbps	54 Mbps	54 Mbps
Range	300 m outdoor	100 m outdoor	300 m outdoor
Modulation	CCK	OFDM	CCK + OFDM
Interference	High (ISM)	Lower (less used)	High (ISM)
Backward compat.	—	Not with 802.11b	With 802.11b

OFDMA vs. MU-MIMO (Wi-Fi 6)

	OFDMA	MU-MIMO
Multiplexing dimension	Frequency	Space
Best for	Many small packets	Large packets, few users
Channel division	Sub-channels (RUs)	Full channel, multiple beams

	OFDMA	MU-MIMO
UL support in 802.11ax	Yes	Yes

PART 4: PRACTICE QUESTIONS

Conceptual:

1. Explain the difference between infrastructure and ad hoc 802.11 networks. What are the architectural implications?
2. Why are there only 3 non-overlapping channels in the 2.4 GHz band? Calculate this from the channel bandwidth and spacing.
3. Explain why CSMA/CA uses "collision avoidance" rather than "collision detection". What makes CD impossible in wireless?
4. Describe the hidden terminal problem and explain precisely how RTS/CTS solves it (what each station does upon hearing each frame).
5. Why does 802.11n introduce HT-Mixed mode rather than simply using Greenfield everywhere?
6. Explain why EDCA still cannot guarantee QoS for a real-time video stream, even though voice is prioritized over background traffic.
7. What is the purpose of scrambling in the DSSS PHY? Why is it different from CRC?
8. In 802.11 power management, what is the difference between TIM and DTIM? Why do both exist?
9. Explain BSS coloring. Why does using a higher CCA threshold for different-color BSSs enable spatial reuse without causing collisions?
10. A station sets its NAV when hearing an RTS. When is the NAV released, and what prevents it from transmitting before then?

Calculation exercises:

1. For 802.11a with CSMA/CA and no RTS/CTS: calculate the throughput for 1500-byte IP packets. (Answer: ~30.8 Mbps)
2. Repeat for 500-byte IP packets and explain why throughput is lower. (Answer: ~16.6 Mbps — PHY overhead is the same but useful data is smaller)
3. For 802.11a with RTS/CTS and 1500-byte packets, calculate throughput and compare to ex. 11. Explain the overhead difference. (Answer: ~25.6 Mbps)
4. Calculate the DIFS and PIFS for 802.11a ($T_{\text{SIFS}} = 16 \mu\text{s}$, $T_{\text{slot}} = 9 \mu\text{s}$).
5. In the contention window exponential back-off, if $CW_{\text{min}} = 15$ and two consecutive collisions occur, what is the current CW?

PART 5: EXAM-READY QUICK REFERENCE

PHY mnemonics: - **basic** → bigger channels (HR-DSSS, 11 Mbps, 2.4 GHz) - **advanced** → another band (OFDM, 54 Mbps, 5 GHz) - **good compatibility** → goes 2.4 GHz with OFDM too - **nice MIMO** → next level (HT, up to 600 Mbps, 2.4+5 GHz) - **ax** = excellent efficiency (OFDMA + MU-MIMO + BSS coloring)

IFS priority order (high → low): SIFS → PIFS → DIFS → AIFS[VO] → AIFS[BE]

Back-off freezes when: another station transmits during your back-off count → timer pauses → resumes when medium idle (fairness mechanism)

RTS/CTS duration chain: - $RTS.duration = T_CTS + T_data + T_ACK + 3 \times SIFS$ - $CTS.duration = T_data + T_ACK + 2 \times SIFS$ - $DATA.duration = T_ACK + SIFS$ (0 if no more fragments)

Aggregation rule of thumb: A-MPDU gives slightly less throughput than A-MSDU because MPDUs each have their own MAC+LLC headers, while A-MSDU subframes only add small subframe headers. However, A-MPDU allows partial retransmission.

802.11ax key insight: Previous Wi-Fi generations increased the pipe (speed). Wi-Fi 6 improves how multiple users share the pipe (efficiency), which is what matters in dense environments like stadiums, offices, and apartments.